EFY Note
The source code of this project is available for free download at *source.efymag.com*

from the supplied air. Also, it removes moisture from the air. From the filter unit, air flow is divided into two parts and supplied to port A of solenoid valves V1 and V2.

During operation, one of the solenoids will be turned on and the other will be turned off. Port B of both valves is the outlet, which is split into two parts. One part is connected to port A of solenoid valve V3 for V1 and port A of solenoid valve V4 for V2. The other is connected to CAN1 for V1 and CAN2 for V2.

When CAN1 is in forward bias, where it is generating oxygen, valve V3 is tuned off and the flow of air is supplied to CAN1. Some part of the air is supplied to CAN2 outlet and that is drained from valve V4, which is turned on at that time and simultaneously valve V2 is turned off. At the same time, valve V5 is turned on to allow the purified oxygen to exit from its port B. Since normally-open relays are used, these get triggered to trigger the coils of valves as per above sequence. The same cycle is repeated, once preprogrammed cycle time is achieved for CAN2.

For better safety, you can use normally-open valves by making slight changes in the program. Thus, the air would exit the system if sudden power failure occurs.

For using this oxygen concentrator at optimum level, you should check flow rate and purity of the output oxygen and change cycle time in the program as per requirement. This is how you can make an oxygen concentrator at home. Note that purity and flow rate is to be verified and monitored continuously.

After uploading the source code to Arduino and wiring done as per circuit diagram, switch on the power supply. Relays RL1, RL4, and RL5 will energise for 40 seconds to open the valves V1, V4, and V5. Similarly, relays RL2, RL3, and RL6 will energise for 40 seconds to open valves V2, V3, and V6. This cycle will repeat again and again until the power is switched off.

Note. EFY lab has tested the control circuit given in Fig. 1. The complete hardware assembly including piping connections, filtration unit, and compressor were tested by the author. The author's prototype with air compressor is shown in Fig. 7.

Caution. The project described here is just a DIY hobby project successfully demonstrated by the author. However, the author and EFY are not liable for any mishap or loss that may occur due to failure of this system while using it. EFY

Tej Vijaykumar Patel is vice president of Non Woven Division, SKAPS Industries India Pvt Ltd, Mundra. His interests include robotics, automation, and product development